Could an invalid-curve weakness distinguish the eleven omissions?

Date: 2026-10-23 UTC. **Status:** Exact check complete for eleven omitted candidates and eight published controls over the same four field primes. This is a conditional analysis of unvalidated point input. The historical program's rejection criterion remains unknown.

Result

Under the usual affine short-Weierstrass addition formulas, **the configured coefficient `b` does not determine which alternate curves can be processed when input points are not checked against the configured curve equation**. For a fixed field prime `p` and coefficient `a`, points on any nonsingular $E_{b'}: y^2=x^3+ax+b'$ are operated on by the same formulas. The 19 exact archived candidates all have `a=-3` and nonzero configured `b`; within each of the four fields they therefore share this conditional invalid-curve surface. A weakness based solely on feeding points from one of these alternate curves into that kind of implementation cannot explain why an otherwise identical implementation would be safe on the published `b` and unsafe on an omitted `b`.

This is an equality of **available wrong-curve inputs under a specified arithmetic model**, not an assertion that a deployed implementation accepts them or that a particular secret-key protocol leaks useful information. The EFD's short-Weierstrass formulas and Antipa et al.'s public-key-validation analysis describe the underlying arithmetic and attack context (`retrieved`, primary). The exact candidate coefficients and checks are here (`internal`, exact computation); Sage separately confirmed every order-two witness (`internal`, separately checked).

Field	Omitted seed ordinals	Published seed ordinals
192 bits	2198, 5405	139, 6013, 5228
239 bits	36, 1562, 2491, 2682, 2863	2185, 2320, 3048
224 bits	777, 1485, 5736	7026
256 bits	876	1842

Every row shares `p,a` within its field and the witness $E_0: y^2=x^3-3x$, $Q=(0,0)$ over that field.

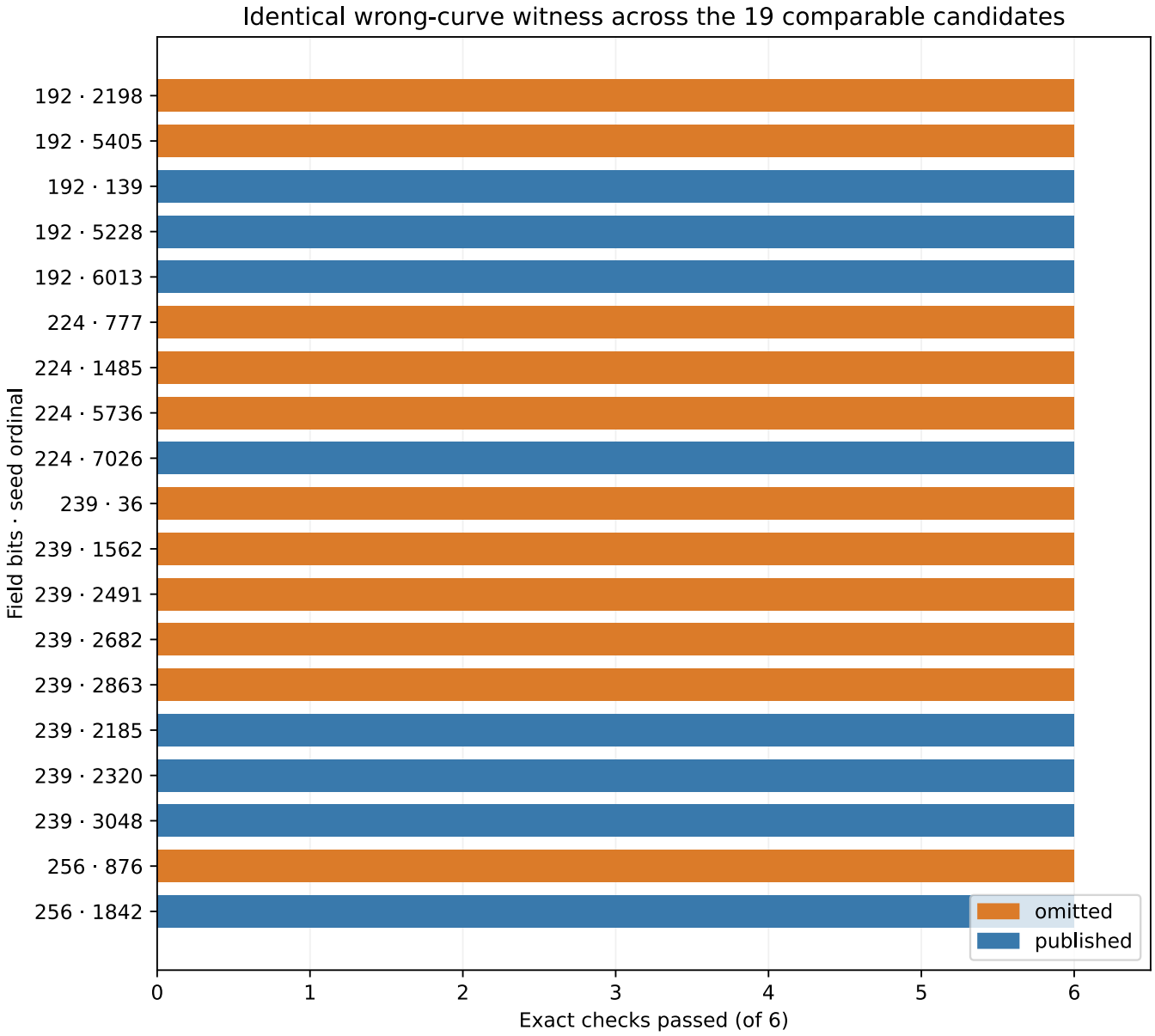


Figure 1: Exact per-candidate check

Figure 1. Every bar is one exact archived candidate. Orange denotes an omitted curve, blue a published control. Each bar reaches all six checks: $a=-3$, $b\neq 0$, nonsingular E_0, $Q\in E_0$, $Q\notin E_b$, and $2Q=O$. Editable source: draw.py.

Why the result follows

For distinct points, the affine slope is $(y_2-y_1)/(x_2-x_1)$; for doubling it is $(3x_1^2+a)/(2y_1)$. The output coordinates are $x_3=\lambda^2-x_1-x_2$ and $y_3=\lambda(x_1-x_3)-y_1$. None of these expressions contains `b`. With the usual identity, inverse and vertical-line cases handled correctly, the formulas form the group law on **whichever** nonsingular `E_b′` contains the input point. If an implementation accepts an input point without first verifying its configured equation, fixing a different `b` does not remove that `E_b′` from the arithmetic it can perform. This algebraic statement does not require that `E_b′` itself be one of the eleven named omitted curves.

There is a simple exact witness on every archived row. With `a=-3`, E_0: $y^2=x^3-3x$ is nonsingular for these field primes. `Q=(0,0)` lies on E_0, is its own inverse, and has order two. Each configured `b` is nonzero, so `Q` fails its equation. The frozen protocol required these checks for all eleven omissions and eight same-field controls. The first implementation verified them against pinned archived JSON SHA-256 digests and retained every `p,a,b`, curve UID and Boolean result. An independent Sage verifier rebuilt each exact curve in `GF(p)`, checked nonsingularity and `Q.order()==2`, and bound its output to the results hash. All 19 passed.

The small-field controls at `p=101` found `[Q,2Q,3Q]=[Q,O,Q]` for two different configured `b` values, and an ordinary nonzero-`y` point on a third curve doubled onto that third curve. These are arithmetic controls, not protocol exploits.

The order-two point can at most distinguish a secret scalar's parity when a protocol exposes the relevant output and the implementation actually accepts this input. A practical invalid-curve key-recovery attack would additionally need points of suitable orders, repeated secret use, observability, and a complete protocol analysis. We have not tested any deployed implementation or secret-key protocol here. For the named eleven omitted curves specifically, the archived group orders are prime, so their *own* full groups do not offer small subgroup factors. Choosing an omitted `b` as an invalid curve against a published configuration therefore does not by itself supply a small-subgroup attack.

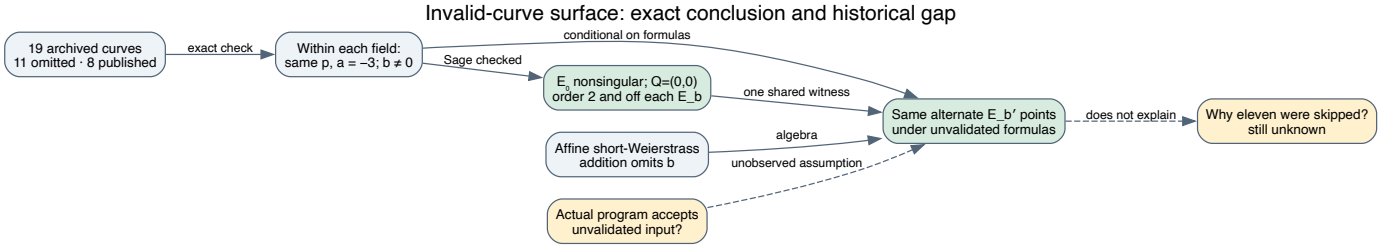


Figure 2: Inference boundary

Figure 2. Solid edges are algebraic implications or exact checks; dashed edges identify assumptions and the still-unobserved historical rejection step. Editable source: inference.dot.

Scope and historical implication

The statement applies to an implementation that uses `b`-independent short-Weierstrass formulas on attacker-supplied points without equation validation. It does not cover `b`-dependent complete formulas, validation and subgroup checks, x-only ladders, faults, fixed-base-only operations, or a protocol that reveals no secret-dependent result. Those choices may create implementation differences, but no such difference has been tied to the eleven historical keep/skip decisions.

Under the user's working premise that the program reached all eleven candidates and rejected them for a hidden weakness, this study rejects one simple version of that premise: **the omitted `b` alone creates a unique invalid-curve input surface while the published `b` avoids it**. The remaining space includes a property of the honest prime-order groups, an implementation or workflow criterion with specified code, an undiscovered construction that exploits other information, or an unrecorded historical decision. None has been demonstrated. Anthropic's reconstruction (`retrieved`, primary) recovers the timestamp-seeded sequence but reports no original acceptance program or complete log. The selection-rule problem stays open. The canonical ECDLP graphs were checked; this conditional input-validation observation creates no verified attack edge.

That premise also has a useful counting consequence: a deterministic hidden weakness that exactly explains the named decisions would mark **all eleven of the nineteen comparable qualifying candidates** as weak and none of the eight published controls. This is a requirement on those observed prefixes, not an estimate of the weakness's prevalence among all curves or all timestamps. An explanation based on an extremely rare special curve must account for this concentration, the unobserved number of timestamp trials, and the two singleton published fields separately.

Reproduction and provenance

Run `python3 research/nist_seed_invalid_curve_surface_20261023/check.py` only in a fresh copy with no existing `results.json`; it refuses to overwrite a record. Run `sage -python research/nist_seed_invalid_curve_surface_20261023/verify.py` only where no `verification.json` exists. The raw `results.json` records a clean source commit `32df781f76be8ef1d0d11f1c4ce2d8f8d4cd8bf3`, host and Python version; `verification.json` binds the exact result and verifier source hashes. Source labels: `internal` for archived candidate data, `derived` for the formula implication, `exact computation` for the 19 rows, and `separately checked` for Sage. No official hypothesis or goal status changes here.